

1. Administrative & Study Identification

Full Study Title: Study of Pembrolizumab (MK-3475) vs Placebo for Participants With Non-small Cell Lung Cancer After Resection With or Without Standard Adjuvant Therapy (MK-3475-091/KEYNOTE-091)
Short Title/Acronym: PEARLS / KEYNOTE-091 **Protocol Number:** MK-3475-091 **NCT Number:** NCT02504372 **Posting ID:** 794471016954218372 **Sponsor Name:** Merck / Merck Sharp & Dohme LLC (MSD) **Investigational Product:** Pembrolizumab (MK-3475 / KEYTRUDA®) **Phase of Development:** Phase 3 (PHASE3) **Trial Status:** Active, Not Recruiting (ACTIVE_NOT_RECRUITING) **Disease Area:** Non-small Cell Lung Cancer **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether pembrolizumab provides improved disease-free survival (DFS) versus placebo in the overall population of participants with early-stage non-small cell lung cancer (NSCLC) after complete surgical resection, and to evaluate DFS in Programmed Death Ligand-1 (PDL-1) strong positive participants with a Tumor Proportion Score (TPS) $\geq 50\%$. **Secondary Objectives:** To evaluate Overall Survival (OS), OS in PDL-1 strong positive participants (TPS $\geq 50\%$), DFS in PDL-1 strong positive participants (TPS $\geq 1\%$), and assess the safety and tolerability profile of pembrolizumab.

2.2 Background & Rationale The benefit of adjuvant chemotherapy in NSCLC has historically plateaued at around a 5.4% improvement in 5-year overall survival and is hampered by treatment toxicities. Immunotherapy approaches, such as anti-PD-1 inhibitors like pembrolizumab, are hypothesized to work best in the context of minimal residual disease. Therefore, the adjuvant setting is considered the ideal clinical scenario to minimize disease recurrence and potentially improve the cure rate after surgery for early-stage NSCLC.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Triple-blind (Participant, Care Provider, Investigator) **Randomization:** 1:1 (Pembrolizumab vs. Placebo), stratified by receipt of adjuvant chemotherapy and region of the world.

3.2 Planned Enrollment & Duration Targeted Enrollment: 1,177 participants (randomized) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** August 2015 **Estimated Primary Completion:** January 2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Pathological diagnosis of Union for International Cancer Control (UICC) v7 Stage IB (≥ 4 cm), II, or IIIA NSCLC confirmed at surgery, any histology. 2. Complete surgical resection (lobectomy or pneumonectomy) with resection margins proved microscopically free of disease (R0). 3. Available tumor sample obtained at surgical resection for programmed cell death ligand-1 (PD-L1) Immunohistochemistry (IHC) expression assessment, and an Eastern Cooperative Oncology Group (ECOG) Performance Status of 0-1.

4.2 Exclusion Criteria 1. Participants with two synchronous primary non-small cell lung cancers, or evidence of disease at clinical/radiological baseline assessment. 2. Prior neoadjuvant radiotherapy or neoadjuvant chemotherapy for the current NSCLC diagnosis, or prior treatment with anti-PD-1, anti-PD-L1/L2, anti-CD137, or CTLA-4 modulators. 3. Pregnant or breastfeeding females (requires negative serum or urine pregnancy test at screening). 4. History of Human Immunodeficiency Virus (HIV), active Hepatitis B or C, or active autoimmune disease requiring systemic treatment in the past 2 years. 5. Administration of a live vaccine within 30 days prior to the first infusion.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product:

Pembrolizumab **Trade Name:** Keytruda **ATC Code:** L01FF02 **EMA Product Information:** [EMA Keytruda Product Information](#) **Dosage/Frequency:** Pembrolizumab 200 mg or Placebo every 3 weeks (Q3W). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 1 year (expected maximum of 18 doses) or until disease recurrence or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for comorbidities. Prior adjuvant platinum-based chemotherapy following complete resection was permitted and received by 86% of randomized patients. **Prohibited:** Concurrent investigational systemic anticancer therapies, chronic use of immunosuppressive agents/systemic corticosteroids, and live vaccines during the treatment period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor sample collection for PD-L1, ECOG assessment, Baseline Imaging
Baseline	Day 0	Randomization, First IV Infusion (Pembrolizumab or Placebo)
Interim	Every 3 Weeks	IV Infusion Administration, Safety Labs, AE monitoring (e.g., pneumonitis, myocarditis)
End of Study	1 Year (18 Doses)	Final Treatment, Transition to Survival and Recurrence Follow-up Phase

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- Disease-Free Survival (DFS):** Defined as the time from randomization to either the date of disease recurrence (local, regional, or distant), occurrence of a second primary cancer, or death from any cause as assessed by the investigator.
- DFS in Programmed Death Ligand-1 (PDL-1) Strong Positive Participants With Tumor Proportion Score (TPS) $\geq 50\%$:** Defined as the time from randomization to either the date of disease recurrence or death from any cause.

7.2 Secondary Outcomes

- DFS in PDL-1 Strong Positive Participants With TPS $\geq 1\%$**
- Overall Survival (OS)**
- OS in PDL-1 Strong Positive Participants With TPS $\geq 50\%$**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of clinical and laboratory AEs, with special attention to immune-mediated adverse reactions such as pneumonitis, myocarditis, hypothyroidism, and hyperthyroidism.

Serious Adverse Events (SAEs): Reported within 24 hours of site awareness, particularly fatal adverse events (e.g., severe myocarditis).

Data Monitoring Committee (DMC): External Independent Data Monitoring Committee (eIDMC) utilized to oversee safety and efficacy during periodic interim analyses.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy outcomes. Safety Set for toxicity evaluations.

Sample Size Justification: Powered at 90% to detect a target treatment effect (Hazard Ratio) for DFS in the overall population and the PD-L1 high expression subgroup.

Handling of Missing Data: Censoring applied at the time of the last adequate disease assessment for DFS.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular on-site and remote monitoring visits to ensure protocol compliance and verify source data. **Audit Trail:**

Robust electronic Case Report Form (eCRF) locking and rigorous data clarification processes managed by the sponsor.

11. Study Identifiers & Governance

Primary ID: MK-3475-091 **Posting ID:** 794471016954218372 **Registry Number:** NCT02504372 **Sponsor/Lead Organization:** Merck / Merck Sharp & Dohme LLC **Study Title:** PEARLS / KEYNOTE-091 **Official Regulatory Title:** A Randomized, Phase 3 Trial With Anti-PD-1 Monoclonal Antibody Pembrolizumab (MK-3475) Versus Placebo for Patients With Early Stage NSCLC After Resection and Completion of Standard Adjuvant Therapy (PEARLS)

12. Clinical Framework (NSCLC Specific)

Primary Condition: Non-small Cell Lung Cancer (UMLS Preferred Name: Non-Small Cell Lung Carcinoma) **Related Disease**

Classification: Metastatic non-small cell lung cancer (recurrent settings) **Diagnosis Threshold:** Stage IB (≥ 4 cm), II, or IIIA NSCLC **Medical Methodology:** Complete Surgical Resection (R0) optionally followed by Adjuvant Platinum-Based Chemotherapy

Optimization Target: Prevention of disease recurrence and improvement of Disease-Free Survival (DFS)

13. Study Procedures & Methodology

Management Pathway: Adjuvant Immunotherapy Clinical Pathway

Education Protocol (HCP): Identification and management protocols for immune-related adverse events (irAEs) **Education Protocol**

(Patient): Patient education on early recognition and reporting of immune-mediated symptoms (e.g., pneumonitis, myocarditis)

Quality Audit Mechanism: Central imaging review and routine source data verification for recurrence endpoints

14. Observation & Follow-Up Schedule

Total Duration: Up to 1 year of active treatment followed by long-term survival follow-up **Onsite Visit Cadence:** Every 3 weeks during the active 1-year treatment phase **Remote Monitoring**

Frequency: Regular survival status checks post-treatment

Checklist Review Interval: Radiological tumor assessments performed every 12 weeks (Q12W) to monitor for recurrence

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					